

Linear Algebra Assignment 11

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Exercise 4.9

Statement

Let $A, B \in \mathbb{K}^{n \times n}$ be two matrices and let $p \leq n$. The two matrices are defined by blocks as

$$A = \begin{pmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{pmatrix}, \quad B = \begin{pmatrix} B_{1,1} & B_{1,2} \\ B_{2,1} & B_{2,2} \end{pmatrix},$$

with

$$A_{1,1} \in \mathbb{K}^{p \times p}, \quad A_{1,2} \in \mathbb{K}^{p \times (n-p)}, \quad A_{2,1} \in \mathbb{K}^{(n-p) \times p}, \quad A_{2,2} \in \mathbb{K}^{(n-p) \times (n-p)}.$$

Same decomposition seem to hold for B .

(i) Prove that

$$AB = \begin{pmatrix} A_{1,1}B_{1,1} + A_{1,2}B_{2,1} & A_{1,1}B_{1,2} + A_{1,2}B_{2,2} \\ A_{2,1}B_{1,1} + A_{2,2}B_{2,1} & A_{2,1}B_{1,2} + A_{2,2}B_{2,2} \end{pmatrix}.$$

(ii) Show that, if $A_{2,1} = 0$, then

$$\det A = \det(A_{1,1}) \det(A_{2,2}).$$

(iii) Show that, if $\det A_{1,1} \neq 0$, then

$$\det A = \det(A_{1,1}) \det(A_{2,2} - A_{2,1}A_{1,1}^{-1}A_{1,2}).$$

Indication: The matrix

$$M = \begin{pmatrix} \text{Id}_p & 0 \\ -A_{2,1}A_{1,1}^{-1} & \text{Id}_{n-p} \end{pmatrix}$$

may be of help.

Solution

Let $A, B \in \mathbb{K}^{n \times n}$, $p \leq n$. Two matrices are defined by blocks as

$$A = \begin{pmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{pmatrix}, \quad B = \begin{pmatrix} B_{1,1} & B_{1,2} \\ B_{2,1} & B_{2,2} \end{pmatrix}.$$

With

$$A_{1,1} \in \mathbb{K}^{p \times p}, \quad A_{1,2} \in \mathbb{K}^{p \times (n-p)}, \\ A_{2,1} \in \mathbb{K}^{(n-p) \times p}, \quad A_{2,2} \in \mathbb{K}^{(n-p) \times (n-p)}.$$

Same decomposition seem to hold for B .

(i) **Prove that**

$$\begin{aligned} AB &= \begin{pmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \end{pmatrix} \begin{pmatrix} B_{1,1} & B_{1,2} \\ B_{2,1} & B_{2,2} \end{pmatrix} \\ &= \begin{pmatrix} A_{1,1}B_{1,1} + A_{1,2}B_{2,1} & A_{1,1}B_{1,2} + A_{1,2}B_{2,2} \\ A_{2,1}B_{1,1} + A_{2,2}B_{2,1} & A_{2,1}B_{1,2} + A_{2,2}B_{2,2} \end{pmatrix}. \end{aligned}$$

Now check dimensions:

$$\dim(A_{1,1}B_{1,1} + A_{1,2}B_{2,1}) = (p \times p)(p \times p) + (p \times (n-p))((n-p) \times p) = p \times p.$$

$$\dim(A_{1,1}B_{1,2} + A_{1,2}B_{2,2}) = (p \times p)(p \times (n-p)) + (p \times (n-p))((n-p) \times (n-p)) = p \times (n-p).$$

$$\dim(A_{2,1}B_{1,1} + A_{2,2}B_{2,1}) = ((n-p) \times p)(p \times p) + ((n-p) \times (n-p))((n-p) \times p) = (n-p) \times p.$$

$$\dim(A_{2,1}B_{1,2} + A_{2,2}B_{2,2}) = ((n-p) \times p)(p \times (n-p)) + ((n-p) \times (n-p))((n-p) \times (n-p)) = (n-p) \times (n-p).$$

As we can see, dimensions match, and $AB \in \mathbb{K}^{n \times n}$.

(ii) **Show that, if $A_{2,1} = 0$, then**

$$\det A = \det(A_{1,1}) \det(A_{2,2}).$$

If $A_{2,1} = 0$, then

$$A = \begin{pmatrix} A_{1,1} & A_{1,2} \\ 0 & A_{2,2} \end{pmatrix}.$$

By the Leibniz formula,

$$\det A = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^n a_{i,\sigma(i)}.$$

Since $\dim(A_{2,1}) = (n-p) \times p$, for rows $i > p$, if $\sigma(i) \leq p$, then

$$a_{i,\sigma(i)} = 0.$$

So, only permutations that will survive should map all rows $p+1, \dots, n$ to columns $p+1, \dots, n$, i.e.

$$\sigma(\{p+1, \dots, n\}) = \{p+1, \dots, n\}$$

and in such case rows $1, \dots, p$ will be mapped to columns $1, \dots, p$, i.e.

$$\sigma(\{1, \dots, p\}) = \{1, \dots, p\}.$$

Then we split $\sigma = (\sigma_1, \sigma_2)$, and

$$\varepsilon(\sigma) = \varepsilon(\sigma_1)\varepsilon(\sigma_2).$$

Hence

$$\sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{i=1}^n a_{i,\sigma(i)} = \left(\sum_{\sigma_1 \in S_p} \varepsilon(\sigma_1) \prod_{i=1}^p a_{i,\sigma_1(i)} \right) \left(\sum_{\sigma_2 \in S_{n-p}} \varepsilon(\sigma_2) \prod_{s=p+1}^n a_{s,\sigma_2(s)} \right).$$

The first sum is $\det(A_{1,1})$, and the second one is $\det(A_{2,2})$. Hence

$$\det(A) = \det(A_{1,1}) \det(A_{2,2}).$$

(iii) Show that, if $\det A_{1,1} \neq 0$,

$$\det A = \det(A_{1,1}) \det(A_{2,2} - A_{2,1}A_{1,1}^{-1}A_{1,2}).$$

Let

$$M = \begin{pmatrix} \text{Id}_p & 0 \\ -A_{2,1}A_{1,1}^{-1} & \text{Id}_{n-p} \end{pmatrix}.$$

Since

$$\det(M) = \det(M^T),$$

and M^T is block upper triangular, by previous point

$$\det(M) = \det(\text{Id}_p) \det(\text{Id}_{n-p}) = 1.$$

Let us multiply $M \cdot A$. By lemma about matrix product,

$$\det(MA) = \det(M) \det(A) = \det(A).$$

Then by point (i)

$$\begin{aligned} MA &= \begin{pmatrix} A_{1,1} & A_{1,2} \\ -A_{2,1}A_{1,1}^{-1}A_{1,1} + A_{2,1} & -A_{2,1}A_{1,1}^{-1}A_{1,2} + A_{2,2} \end{pmatrix} \\ &= \begin{pmatrix} A_{1,1} & A_{1,2} \\ 0 & A_{2,2} - A_{2,1}A_{1,1}^{-1}A_{1,2} \end{pmatrix}. \end{aligned}$$

By point (ii),

$$\det(MA) = \det(A_{1,1}) \det(A_{2,2} - A_{2,1}A_{1,1}^{-1}A_{1,2}) = \det(A),$$

proving desired property.

Exercise 4.13

Statement

Let $A \in \mathbb{K}^{n \times n}$ with $n \in \mathbb{N}$, with coefficients a_{ij} , $1 \leq i, j \leq n$. For each pair of indices (i, j) , define

$$C_{ij} = (-1)^{i+j} \det(A_{ij}^*),$$

where A_{ij}^* is the matrix obtained by deleting row i and column j . Then

$$\text{adj}(A) = \begin{pmatrix} C_{11} & \cdots & C_{n1} \\ \vdots & \ddots & \vdots \\ C_{1n} & \cdots & C_{nn} \end{pmatrix}.$$

(i) Prove that

$$A \text{adj}(A) = \det(A)I_n.$$

(ii) Conclude that when A is invertible, we have the following formula for the inverse matrix:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

What is $\det(\text{adj}(A))$?

Solution

Let $A \in \mathbb{K}^{n \times n}$ with $n \in \mathbb{N}$, with a_{ij} , $1 \leq i, j \leq n$. For each pair of indices (i, j) , we define

$$C_{ij} = (-1)^{i+j} \det(A_{ij}^*).$$

Then

$$\text{adj}(A) = \begin{pmatrix} C_{11} & \cdots & C_{n1} \\ \vdots & \ddots & \vdots \\ C_{1n} & \cdots & C_{nn} \end{pmatrix}.$$

(i) Let $P = A \cdot \text{adj}(A)$.

$$p_{ii} = \sum_{j=1}^n a_{ij} C_{ij} = \sum_{j=1}^n a_{ij} (-1)^{i+j} \det(A_{ij}^*) = \det A, \quad \forall i \in \{1, \dots, n\}.$$

(ii) $i \neq k$.

$$p_{ik} = \sum_{s=1}^n a_{is} C_{ks} = \sum_{s=1}^n a_{is} (-1)^{s+k} \det(A_{ks}^*).$$

But this is exactly the Laplace expansion of a matrix when $R_i = R_k$. Let

$$B = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{i1} & \cdots & a_{in} \\ a_{i1} & \cdots & a_{in} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix},$$

so $R_i = R_k$, i.e. row k replaced by row i . Then

$$\sum_{s=1}^n a_{is} (-1)^{s+k} \det(A_{ks}^*) = \sum_{s=1}^n b_{ks} (-1)^{s+k} \det(B_{ks}^*) = \det(B) = 0$$

by alternating property of determinant. Hence

$$p_{ik} = 0 \quad \forall i \neq k.$$

Since

$$A \text{adj}(A) = \det(A) I_n,$$

now $\text{adj}(A)A = \det(A)I_n$ similarly by the same logic as in previous point.

If $\det A \neq 0$, then

$$A \cdot \frac{1}{\det A} \text{adj}(A) = \frac{1}{\det A} \det(A) I = I.$$

Hence

$$A^{-1} = \frac{1}{\det A} \text{adj}(A).$$

Then

$$\text{adj}(A) = A^{-1} \det(A).$$

Therefore

$$\det(\text{adj}(A)) = \det(A^{-1} \det(A)) = (\det A)^n \det(A^{-1}) = \frac{(\det A)^n}{\det A} = (\det A)^{n-1}.$$

Exercise 5.3

Statement

Exercise 5.3. (Shift of spectrum)

Let $A \in \mathbb{K}^{n \times n}$ and $\alpha \in \mathbb{K}$. Show that $\alpha I_n + A$ is diagonalizable if and only if A is diagonalizable. What are the eigenvalues and associated eigenspaces of $\alpha I_n + A$?

Solution

$\Rightarrow$ Assume A is diagonalizable. Then there exists a basis E made of eigenvectors of A ,

$$E = \{e_1, \dots, e_n\},$$

and eigenvalues $\lambda_1, \dots, \lambda_n$, not necessarily distinct. Then consider

$$(A + \alpha I_n)e_i = Ae_i + \alpha e_i = \lambda_i e_i + \alpha e_i = (\lambda_i + \alpha)e_i.$$

Then $e_1, \dots, e_n$ are eigenvectors of $A + \alpha I_n$, and since those vectors are linearly independent, they form an eigenbasis of $A + \alpha I_n$, with eigenvalues

$$\lambda_1 + \alpha, \dots, \lambda_n + \alpha.$$

Hence $A + \alpha I_n$ is diagonalizable.

$\Leftarrow$ Assume $A + \alpha I_n$ is diagonalizable. Then let

$$B = A + \alpha I_n.$$

B is diagonalizable. Let $b = -\alpha$. Then

$$B + bI_n = A.$$

So we reduced our problem to previous implication. Hence A is diagonalizable.

The eigenspaces are the same, only the eigenvalues shift:

$$V_\lambda^A = V_{\lambda+\alpha}^{A+\alpha I_n}.$$

Exercise 5.5

Statement

Let A be the $n \times n$ matrix defined by

$$A = \begin{pmatrix} a & 1 & 1 & \cdots & 1 \\ 1 & a & 1 & \cdots & 1 \\ 1 & 1 & a & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & a \end{pmatrix}.$$

Is A diagonalizable over $\mathbb{K}$? If so, obtain the diagonalized form. Use the result of Exercise 4.14 and Exercise 5.3.

Solution

Using the result of ex. 4.14, consider

$$B = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 1 & 1 & \cdots & 1 \\ 1 & 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & 1 & \cdots & 1 \end{pmatrix}.$$

Then

$$A = B + (a - 1)I.$$

Using exercise 4.14 result, i.e.

$$\det(w - \lambda e_1, \dots, w - \lambda e_n) = (-\lambda)^n + s(-\lambda)^{n-1}, \quad s = \sum_{i=1}^n w_i,$$

where here $w = (1, \dots, 1)^T$ and $s = n$, hence applying this result to our problem,

$$\det(B - \lambda I) = (-\lambda)^n + n(-\lambda)^{n-1}.$$

Up to the same sign convention for the characteristic polynomial, this gives the roots

$$\lambda = n, \quad \lambda = 0.$$

So, eigenvalues are

$$\lambda_1 = n, \quad \lambda_2 = 0,$$

with

$$m_n = 1, \quad m_0 = n - 1.$$

V_0^B is exactly the nullity of our matrix. Then by rank-nullity theorem

$$\dim(V_0^B) = n - 1,$$

and

$$V_n^B = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \right\} \Rightarrow \dim(V_n^B) = 1.$$

Hence algebraic multiplicities match geometric ones. Hence B is diagonalizable, with

$$D_B = \begin{pmatrix} n & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}.$$

By ex. 5.3,

$$A = B + (a - 1)I,$$

and eigenvalues of $A = B + (a - 1)I$ are

$$\mu_1 = n + (a - 1)$$

with

$$m_{n+a-1} = 1,$$

and

$$\mu_2 = a - 1$$

with

$$m_{a-1} = n - 1.$$

Then

$$D_A = \begin{pmatrix} n+a-1 & 0 & \cdots & 0 \\ 0 & a-1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a-1 \end{pmatrix}.$$